BRAC UNIVERSITY **Set:A**

Department of Computer Science and Engineering

Examination: Mid Term Semester: FALL 2025

Duration: 1 Hour 20 Minutes Full Marks: 20

CSE 420: Compiler Design

Answer the following questions.Theoretical questions should be answered in a precise, concise, and to-the-point manner. Figures in the right margin indicate marks.

Name:	ID:	Section:

1. CO3 For a production rule of the form $S \rightarrow \alpha X \beta$, it is stated that FOLLOW(X)=FIRST(β) and if $\beta \Rightarrow \varepsilon$ then FOLLOW(X)=FOLLOW(S). Explain clearly why this property holds in context-free grammar analysis and how it relates to the computation of FOLLOW sets during parsing. 3

2. CO3 Show how we can detect shift-reduce conflict from automaton states with an example? Explain why this may not be a problem for a SLR parser? 2

3. CO3 Check the Non-Augmented CFG: 5

1. $S \rightarrow A\ S$
2. $S \rightarrow \epsilon$
3. $A \rightarrow W$
4. $A \rightarrow$ other
5. $W \rightarrow$ while E do S end
6. $E \rightarrow$ id

Terminals are: {other, while, do, end, id}

The Parsing table is given below:

State	Action						Go To			
	other	while	do	end	id	$	S	A	W	E
0	S4	S5		R2		R2	1	2	3	
1						Accepted				
2	S4	S5		R2		R2	6	2	3	
3	R3	R3		R3		R3				
4	R4	R4		R4		R4				
5					S7					8
6				R1		R1				
7			R6							
8			S9							
9	S4	S5		R2		R2	10	2	3	
10				S11						
11	R5	R5		R5		R5				

Determine by simulating parsing whether the following string is syntactically correct or not according to the above parsing table. The space in the string is separating the tokens received from the lexical analyzer.

Input String : ***while id do end***

4. What is panic mode error recovery? Explain why panic mode recovery may lead to the compiler finding more errors than that actually exist in the source code. 3
CO1

5. Explain how the structure of the parsing table should change if we have an LR(3) parser as opposed to an LR(1) parser. 2
CO3

6. You are provided with an abstract syntax tree and a set of productions of a grammar – for the input sentence ***x + ((y + z) * w)***. You have to fill in the missing rules so that your semantic rules match the ordering of the nodes in the parse tree. 5
CO5

$E \rightarrow E_1 + T$ *{Write your rules here}*

$E \rightarrow T$ {E.node = T.node;}

$T \rightarrow T_1 * F$ *{Write your rules here}*

$T \rightarrow F$ {T.node = F.node}

$F \rightarrow id$ *{Write your rules here}*

$F \rightarrow (E)$ {F.node = E.node}